Lecture 1: Practice Problems

Linear Algebra

November 14, 2026

1 Number Sets and Fields

Problem 1.1. For each of the numbers

$$\frac{7}{3}, \qquad \sqrt{49}, \qquad \sqrt{12}, \qquad 0.25, \qquad \pi - \pi, \qquad (1+i)(1-i), \qquad 2+3i,$$

decide which of the sets $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$ it belongs to.

Problem 1.2 (**Mandatory**)**.** Each of the following sets, with the usual addition and multiplication, is not a field. For each set, find a field axiom that fails.

(a) $\mathbb{N} = \{0, 1, 2, \ldots\}$;

(b) the set $\mathbb{R}_{>0}$ of positive real numbers;

(c) the set $\{0, 1\}$;

(d) the set $2\mathbb{Z} = \{\ldots, -4, -2, 0, 2, 4, \ldots\}$ of even integers;

(e) the set $\{0\}$.

Problem 1.3. (a) Show that $\mathbb{Q}(\sqrt{2}) = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$, with the usual addition and multiplication, is a field. (Hint: to find the inverse of $a + b\sqrt{2} \neq 0$, multiply by $\frac{a-b\sqrt{2}}{a-b\sqrt{2}}$. Why is $a^2 - 2b^2 \neq 0$?)

(b) Is $\{a + b\sqrt{2} : a, b \in \mathbb{Z}\}$ a field?

2 Vector Spaces

Problem 2.1 (**Mandatory**)**.** On $\mathbb{R}^2$, keep the usual vector addition, but define a new scalar multiplication. In each case, decide which vector space axioms hold and which fail.

(a) $c \odot (x, y) = (cx, 0)$;

(b) $c \odot (x, y) = (cx, y)$.

Problem 2.2. On the set $V = \mathbb{R}_{>0}$ of positive real numbers, define "addition" and "scalar multiplication" by

$$u \oplus v = uv \qquad \text{and} \qquad c \odot u = u^c \qquad (u, v \in V,\ c \in \mathbb{R}).$$

Show that V is a vector space over $\mathbb{R}$. What is its zero vector, and what is the additive inverse of u?

Problem 2.3. (a) Show that the set of all functions $f : \mathbb{R} \to \mathbb{R}$, with $(f+g)(x) = f(x) + g(x)$ and $(cf)(x) = cf(x)$, is a vector space over $\mathbb{R}$. What is its zero vector?

(b) Is the set of all functions $f : \mathbb{R} \to \mathbb{R}$ with $f(0) = 1$, with the same operations, a vector space?

Problem 2.4 (**Mandatory**)**.** For each function, find the largest n such that it lies in $C^n(\mathbb{R})$ (write $n = \infty$ for smooth functions and $n = 0$ for $C(\mathbb{R})$), or say that it is not continuous on $\mathbb{R}$.

(a) e^{-x^2};

(b) $|x|^3$;

(c) $\sqrt{|x|}$;

(d) $f(x) = \frac{\sin x}{x}$ for $x \neq 0$, and $f(0) = 1$;

(e) $g(x) = 1$ for $x \geq 0$, and $g(x) = 0$ for $x < 0$.

Problem 2.5. Which of the following subsets of $C(\mathbb{R})$ are vector spaces with the pointwise operations? Justify your answers.

(a) The functions with $f(x+1) = f(x)$ for all x (the periodic functions with period 1).

(b) The increasing functions, i.e., those with $f(x) \leq f(y)$ whenever $x \leq y$.

(c) The functions with $\lim_{x\to\infty} f(x) = 0$.

(d) The functions with $|f(x)| \leq 1$ for all x.

(e) The functions that are 0 outside the interval $[0, 1]$.

3 Bases and Dimension

Problem 3.1 (**Mandatory**)**.** Decide whether the vectors $(1, 2, 1)$, $(2, 1, 0)$ and $(1, -1, -1)$ in $\mathbb{R}^3$ are linearly independent. If they are not, find a nontrivial linear combination that equals 0, and find a basis and the dimension of their span.

Problem 3.2 (**Mandatory**)**.** One can check that $U = \{(x, y, z, w) \in \mathbb{R}^4 : x+y+z+w = 0 \text{ and } x = w\}$ is a vector space. Find a basis of U and its dimension.

Problem 3.3. Show that $\mathcal{B} = (1,\ x-1,\ (x-1)^2)$ is an ordered basis of the vector space $\mathbb{R}[x]_{\leq 2}$ of polynomials of degree at most 2. Find the coordinate vectors $[x^2]_{\mathcal{B}}$ and $[3 + 2x + x^2]_{\mathcal{B}}$.

4 Linear Transformations

Problem 4.1 (**Mandatory**)**.** Which of the following maps are linear transformations? Justify your answers.

(a) $T : \mathbb{R}^2 \to \mathbb{R}^2$, $T(x, y) = (x - y, 3x)$;

(b) $T : \mathbb{R}^2 \to \mathbb{R}^2$, $T(x, y) = (x + y, 1)$;

(c) $T : \mathbb{R}^3 \to \mathbb{R}^2$, $T(x, y, z) = (xy, z)$;

(d) $D : \mathbb{R}[x]_{\leq 3} \to \mathbb{R}[x]_{\leq 3}$, $D(p) = p'$ (the derivative);

(e) $T : \mathbb{R} \to \mathbb{R}$, $T(x) = |x|$;

(f) $T : \mathbb{R}^2 \to \mathbb{R}^2$, $T(x, y) = (y, x)$.

Problem 4.2 (**Mandatory**)**.** Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be defined by $T(x, y, z) = (x + y,\ y + z,\ x - z)$. Find the kernel and the image of T. Is T injective? Is it surjective?

Problem 4.3. For each of the following linear transformations of the plane, compute $T(\mathbf{e}_1)$ and $T(\mathbf{e}_2)$, sketch the image of the unit square and of the letter F from the gallery figure in the lecture, and find the kernel and the image of T.

(a) the reflection in the line $y = x$, $T(x, y) = (y, x)$;

(b) the projection onto the x-axis, $T(x, y) = (x, 0)$;

(c) the shear $T(x, y) = (x + 2y, y)$;

(d) $T(x, y) = (-x, -y)$.

In which cases is the image of the letter F mirrored?

Problem 4.4. Let $D : \mathbb{R}[x]_{\leq 3} \to \mathbb{R}[x]_{\leq 3}$, $D(p) = p'$. Find the kernel and the image of D, and decide whether D is injective or surjective. What is $D \circ D \circ D \circ D$?

Problem 4.5 ($\star$)**.** Let S, T, U be nonempty sets, and let $f : S \to T$ and $g : T \to U$ be functions.

(a) Show that if $g \circ f$ is injective, then f is injective.

(b) Give an example in which $g \circ f$ is injective, but g is not.

(c) Show that if $g \circ f$ is surjective, then g is surjective.

(d) Let $A : \mathbb{R}^m \to \mathbb{R}^n$ and $B : \mathbb{R}^n \to \mathbb{R}^m$ be linear transformations with $B \circ A = \mathrm{id}_{\mathbb{R}^m}$. Which of A and B must be injective, and which must be surjective?

Problem 4.6 ($\star$)**.** Let $\mathcal{F}$ be the set of all functions $\mathbb{R} \to \mathbb{R}$, with pointwise addition $(f + g)(x) = f(x) + g(x)$ and composition $(f \circ g)(x) = f(g(x))$.

(a) Show that $\circ$ is not commutative.

(b) Show that the right distributive law $(g + h) \circ f = (g \circ f) + (h \circ f)$ always holds.

(c) Show that the left distributive law $f \circ (g + h) = (f \circ g) + (f \circ h)$ can fail. (Try $f(x) = x^2$.)

(d) Show that the left distributive law holds whenever f is linear, i.e., $f(x) = cx$ for some $c \in \mathbb{R}$. Compare with the distributive laws for matrix multiplication.

Problem 4.7 (**Mandatory**)**.** (a) Show that the following map is an isomorphism:

$$T : \mathbb{R}[x]_{\leq 3} \to \mathbb{R}^4, \qquad T(p) = (p(0), p'(0), \tfrac{1}{2}p''(0), \tfrac{1}{6}p'''(0)).$$

(Hint: compute $T(a_0 + a_1x + a_2x^2 + a_3x^3)$.)

(b) Show that the vector space V of Problem 2.2 is isomorphic to $\mathbb{R}$. (Hint: logarithm.)

(c) Which of the following pairs of real vector spaces are isomorphic: $\mathbb{R}[x]_{\leq 5}$ and $\mathbb{R}^6$; $\mathbb{C}$ (over $\mathbb{R}$) and $\mathbb{R}^3$; $\mathbb{R}^2 \times \mathbb{R}^3$ and $\mathbb{R}^5$?

Problem 4.8 (**Mandatory**)**.** Which of the following maps are linear transformations? Justify your answers.

(a) $C^2(\mathbb{R}) \to C(\mathbb{R})$, $f \mapsto f'' - 3f' + 2f$;

(b) $C[0,1] \to \mathbb{R}$, $f \mapsto \int_0^1 xf(x)\,dx$;

(c) $C[0,1] \to \mathbb{R}$, $f \mapsto \int_0^1 f(x)^2\,dx$;

(d) $C(\mathbb{R}) \to C(\mathbb{R})$, $f \mapsto g$ with $g(x) = f(x+1)$;

(e) $C^1(\mathbb{R}) \to C(\mathbb{R})$, $f \mapsto f' + 1$;

(f) $C(\mathbb{R}) \to \mathbb{R}$, $f \mapsto f(0)f(1)$.

Problem 4.9. Let $D : C^\infty(\mathbb{R}) \to C^\infty(\mathbb{R})$, $Df = f'$, and $J : C^\infty(\mathbb{R}) \to C^\infty(\mathbb{R})$, $(Jf)(x) = \int_0^x f(t)\,dt$.

(a) Compute $D(Jf)$ and $J(Df)$ for $f(x) = \cos x$ and for $f(x) = e^{2x}$.

(b) Show that $J(Df) = f - f(0)$ for every f, and find all f with $J(Df) = f$.

(c) Show that $D^2 = D \circ D$ is a linear transformation, and that cos and sin lie in the kernel of $D^2 + \mathrm{id}$, i.e., that they satisfy $f'' + f = 0$.

5 Subspaces

Problem 5.1 (**Mandatory**)**.** Which of the following subsets of $\mathbb{R}^3$ are subspaces?

(a) $\{(x,y,z) : x = 2y\}$;

(b) $\{(x,y,z) : z = x^2\}$;

(c) $\{(x,y,z) : x + y = z + 1\}$;

(d) $\{(x,y,z) : x = 0 \text{ or } y = 0\}$;

(e) $\{(t,2t,3t) : t \in \mathbb{R}\}$;

(f) $\{(x,y,z) : x + y + z = 0 \text{ and } x = y\}$.

Problem 5.2. Let U and W be subspaces of a vector space V.

(a) Show that $U \cap W$ is a subspace of V.

(b) Show that $U \cup W$ need not be a subspace.

(c) Show that $U \cup W$ is a subspace if and only if $U \subseteq W$ or $W \subseteq U$.

Problem 5.3. Show that $\{p \in \mathbb{R}[x]_{\leq 3} : p(1) = 0\}$ is a subspace of $\mathbb{R}[x]_{\leq 3}$ by writing it as the kernel of a linear transformation. Find a basis of this subspace and its dimension.

Problem 5.4 (Mandatory). (a) Show that $U = \{f \in C^2(\mathbb{R}) : f'' + f = 0\}$ is a subspace of $C^2(\mathbb{R})$, by writing it as the kernel of a linear transformation. Show that $a\cos x + b\sin x \in U$ for all $a, b \in \mathbb{R}$.

(b) Show that the even functions ($f(-x) = f(x)$ for all x) and the odd functions ($f(-x) = -f(x)$ for all x) form subspaces of $C(\mathbb{R})$, and that every $f \in C(\mathbb{R})$ is the sum of an even and an odd function. (Hint: $f(x) = \frac{f(x)+f(-x)}{2} + \frac{f(x)-f(-x)}{2}$.)

(c) Is $\{f \in C^\infty(\mathbb{R}) : f(0) = f'(0)\}$ a subspace of $C^\infty(\mathbb{R})$? Is the set of polynomials of degree exactly 2 a subspace of $\mathbb{R}[x]$?

6 Matrices

6.1 Matrix Operations

Problem 6.1 (Mandatory). Let $A = \begin{pmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \end{pmatrix}$ and $\mathbf{x} = (2, 1, -1)^\top$.

(a) Compute $A\mathbf{x}$ in two ways: entry by entry, and as a linear combination of the columns of A.

(b) Compute $A^\top$, $AA^\top$ and $A^\top A$. Which of these matrices are symmetric?

Problem 6.2. Let A be a 2×3 matrix, B a 3×3 matrix and C a 3×2 matrix. Which of the following expressions are defined, and what are their sizes?

$$AB, \qquad BA, \qquad AC, \qquad CA, \qquad A + C^\top, \qquad A^\top + C, \qquad ABC.$$

Problem 6.3. Let $R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ be the rotation by 90°, and $F = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ the reflection in the x-axis. Compute FR and RF, and describe both as geometric transformations. What does this tell you about the order in which linear transformations are composed?

Problem 6.4. Let $R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ be the rotation by the angle θ.

(a) Explain geometrically why $R_\alpha R_\beta = R_{\alpha+\beta}$.

(b) Multiply out $R_\alpha R_\beta$, and deduce the addition formulas for $\cos(\alpha + \beta)$ and $\sin(\alpha + \beta)$.

(c) Show that $(R_\theta)^n = R_{n\theta}$, and conclude that $(R_{2\pi/n})^n = I_2$.

Problem 6.5 ($\star$). For a complex number $z = a + bi$, let $M(z) = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$.

(a) Show that $M(z + w) = M(z) + M(w)$ and $M(zw) = M(z)M(w)$ for all $z, w \in \mathbb{C}$.

(b) Show that $M(1 + i) = \sqrt{2}\, R_{\pi/4}$, with R_θ as in Problem 6.4.

(c) Use parts (a) and (b) to compute $(1 + i)^{10}$.

6.2 Row Reduction

Problem 6.6. Which of the following matrices are in row echelon form? Which are in reduced row echelon form?

$$\begin{pmatrix} 1 & 3 & 0 & 2 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 3 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}.$$

Problem 6.7 (**Mandatory**)**.** Find the reduced row echelon form and the rank of

$$P = \begin{pmatrix} 1 & 2 & 1 & 0 \\ 2 & 4 & 0 & 2 \\ 1 & 2 & 3 & -2 \end{pmatrix}.$$

Problem 6.8 (**Mandatory**, $\star$)**.** Consider the system of linear equations

$$\begin{aligned} w + 3x + 2y + 2z &= 0, \\ w + 4x + y &= 0, \\ 3w + 5x + 10y + 14z &= 0, \\ 2w + 5x + 5y + 6z &= 0. \end{aligned}$$

(a) Find all solutions $(w, x, y, z) \in \mathbb{R}^4$ by bringing the coefficient matrix into reduced row echelon form.

(b) Check that $(-5, 1, 1, 0)$ and $(0, 2, -8, 5)$ are solutions. Is every solution a scalar multiple of $(-5, 1, 1, 0)$?

(c) What is the dimension of the space of solutions?

6.3 The Inverse of a Matrix

Problem 6.9. Find the inverse of $A = \begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix}$ by solving the equation $A\mathbf{x} = \mathbf{y}$ for $\mathbf{x}$, as in the lecture. Check your answer by computing AA^{-1}.

Problem 6.10 (**Mandatory**)**.** Use the Gauss–Jordan method to find the inverse of

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix},$$

and use it to solve the system $A\mathbf{x} = (1, 2, 3)^\top$.

Problem 6.11. (a) Apply the Gauss–Jordan method to $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{pmatrix}$ to show that A is singular, and find a nonzero vector $\mathbf{x}$ with $A\mathbf{x} = \mathbf{0}$.

(b) Show more generally that every 3×3 matrix whose rows form an arithmetic progression (i.e., row 2 − row 1 = row 3 − row 2) is singular.

Problem 6.12. Let $A = \begin{pmatrix} 0 & 2 \\ 1 & 3 \end{pmatrix}$. Find elementary matrices E_1, E_2, E_3 such that $E_3E_2E_1A = I_2$. Use them to write A^{-1}, and also A itself, as a product of elementary matrices.

Problem 6.13 ($\star$)**.** A square matrix A with $A^2 = A$ is called idempotent.

(a) Show that the only invertible idempotent matrix is the identity.

(b) Show that $P = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ is idempotent, and find the kernel and the image of T_P. Describe T_P geometrically.

(c) Show that if A is idempotent, then so is $I - A$, and that $A(I - A) = 0$.

(d) Let A be an idempotent $n \times n$ matrix. Show that $\operatorname{im}(T_A) \cap \ker(T_A) = \{\mathbf{0}\}$, and that every $\mathbf{v} \in \mathbb{R}^n$ can be written as $\mathbf{v} = A\mathbf{v} + (\mathbf{v} - A\mathbf{v})$, where $A\mathbf{v} \in \operatorname{im}(T_A)$ and $\mathbf{v} - A\mathbf{v} \in \ker(T_A)$.

Problem 6.14. Let $N = \begin{pmatrix} 0 & 1 & 2 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{pmatrix}$.

(a) Compute N^2 and N^3.

(b) Show that $(I - N)(I + N + N^2) = I$, and compute $(I - N)^{-1}$.

(c) More generally, show that if $N^k = 0$ for some $k \geq 1$, then $I - N$ is invertible, with $(I-N)^{-1} = I + N + N^2 + \cdots + N^{k-1}$. (Compare with the geometric series $\frac{1}{1-x} = 1 + x + x^2 + \cdots$.)

Problem 6.15 (**Mandatory**)**.** Let $T : \mathbb{R}[x]_{\leq 2} \to \mathbb{R}^3$, $T(p) = (p(-1), p(0), p(1))$.

(a) Find the matrix of T with respect to the ordered basis $(1, x, x^2)$ of $\mathbb{R}[x]_{\leq 2}$ and the standard basis of $\mathbb{R}^3$.

(b) Show that this matrix is invertible, and conclude that T is an isomorphism.

(c) Find the polynomial $p \in \mathbb{R}[x]_{\leq 2}$ with $p(-1) = 0$, $p(0) = 1$, $p(1) = 0$, and the polynomial q with $q(-1) = 1$, $q(0) = 2$, $q(1) = 5$.

6.4 Special Types of Matrices

Problem 6.16. Find the dimensions of the following vector spaces, and a basis of each of the first two:

(a) Sym_2;

(b) Skew_2;

(c) $\mathbb{R}^{2\times 3}$;

(d) Sym_4;

(e) Skew_4;

(f) the space of upper triangular 3×3 matrices with trace 0.

Problem 6.17. (a) Which of the following are groups, and which of the groups are abelian? $(\mathbb{Z}, +)$; $(\mathbb{N}, +)$ with $\mathbb{N} = \{0, 1, 2, \ldots\}$; $(\mathbb{Q} \setminus \{0\}, \cdot)$; $(\mathbb{Z} \setminus \{0\}, \cdot)$; the positive real numbers under multiplication; the four complex numbers $\{1, i, -1, -i\}$ under multiplication.

(b) On $\mathbb{R}$, define $a*b = a+b+ab$. Show that $*$ is associative and commutative and has an identity element, but that $(\mathbb{R}, *)$ is not a group. Is $(\mathbb{R}\setminus\{-1\}, *)$ a group? (Hint: $a*b = (1+a)(1+b)-1$.)

(c) Show that in every group, $(ab)^{-1} = b^{-1}a^{-1}$. Check this for invertible matrices.

Problem 6.18. Which of the following sets of matrices are matrix groups, i.e., contain the identity and are closed under multiplication and under taking inverses?

(a) the invertible diagonal $n \times n$ matrices;

(b) the invertible symmetric 2×2 matrices;

(c) the 2×2 matrices A with $A^2 = I_2$;

(d) the upper triangular $n \times n$ matrices whose diagonal entries are all equal to 1;

(e) the $n \times n$ permutation matrices. How many of them are there?

Problem 6.19. Use the Gauss–Jordan method to find the inverse of $\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{pmatrix}$. Then explain why the inverse of every invertible upper triangular matrix is again upper triangular.

Problem 6.20. Find all orthogonal 2×2 matrices. (Hint: for $Q = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the condition $Q^\top Q = I_2$ means $a^2 + c^2 = 1$, $b^2 + d^2 = 1$ and $ab + cd = 0$. Write $(a, c) = (\cos\theta, \sin\theta)$.) Which of them are rotations, and which are reflections?

6.5 Trace and Determinant

Problem 6.21 ($\star$)**.** Let D_n be the determinant of the $n \times n$ matrix with 2s on the diagonal, 1s directly above and below the diagonal, and 0s everywhere else; for example,

$$D_3 = \det \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}.$$

(a) Compute D_1, D_2 and D_3.

(b) Expand along the first row to show that $D_n = 2D_{n-1} - D_{n-2}$ for $n \geq 3$.

(c) Find a formula for D_n.

Problem 6.22 (**Mandatory**)**.** Compute the determinant of

$$A = \begin{pmatrix} 2 & 0 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 0 & 3 & 1 & 1 \\ 1 & 0 & 2 & 1 \end{pmatrix}$$

in two ways: by cofactor expansion along a well-chosen row or column, and by row reduction. Is A invertible?

Problem 6.23 (**Mandatory**)**.** Let A and B be 3×3 matrices with $\det(A) = 2$ and $\det(B) = -3$. Compute

$$\det(AB), \qquad \det(A^\top B^2), \qquad \det(3A), \qquad \det(A^{-1}B), \qquad \det(BAB^{-1}).$$

Let C be obtained from A by swapping rows 1 and 3 and then adding 5 times row 1 to row 2. What is $\det(C)$?

Problem 6.24. (a) Use row operations to show that

$$\det \begin{pmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{pmatrix} = (b-a)(c-a)(c-b).$$

(b) Conclude that for distinct numbers a, b, c and arbitrary values y_1, y_2, y_3, there is exactly one polynomial $p \in \mathbb{R}[x]_{\leq 2}$ with $p(a) = y_1$, $p(b) = y_2$ and $p(c) = y_3$. Compare with Problem 6.15.

Problem 6.25 ($\star$)**.** Let $A = \begin{pmatrix} 3 & 5 & 3 \\ 1 & 7 & 3 \\ 1 & 2 & 8 \end{pmatrix}$.

(a) For which of the numbers $\lambda = 2, 3, 5$ is the matrix $A - \lambda I_3$ singular?

(b) For each such λ, find a nonzero vector $\mathbf{x}$ with $A\mathbf{x} = \lambda\mathbf{x}$.

(c) Without computing any determinant, find another number λ and a nonzero vector $\mathbf{x}$ with $A\mathbf{x} = \lambda\mathbf{x}$. (Hint: add up the entries in each row of A.)

Problem 6.26 ($\star$)**.** Let

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 0 & 2 & 3 & 4 & 5 \\ 0 & 0 & 3 & 4 & 5 \\ 0 & 0 & 0 & 4 & 5 \\ 0 & 0 & 0 & 0 & 5 \end{pmatrix}.$$

(a) Compute $\det(A)$. Is A invertible? Can A be transformed into I_5 by elementary row operations?

(b) Find the last row of A^2 without computing all of A^2.

(c) Explain why $A - I_5$ and $A - 2I_5$ are singular. Find nonzero vectors $\mathbf{x}$ and $\mathbf{y}$ with $A\mathbf{x} = \mathbf{x}$ and $A\mathbf{y} = 2\mathbf{y}$.

Problem 6.27. (a) Find the area of the triangle with vertices $(1, 1)$, $(4, 2)$ and $(2, 5)$. (Hint: move one vertex to the origin.)

(b) Let $T(x, y) = (2x - y, x + 3y)$. Find the area of the image under T of the unit square and of the unit disk.

(c) Show that the image of the unit disk under $T(x, y) = (ax, by)$, with $a, b > 0$, is the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1$, and conclude that its area is πab.

(d) Which of the transformations of Problem 4.3 reverse the orientation? Compare with your sketches of the letter F.

Problem 6.28. The Fibonacci numbers are defined by $F_0 = 0$, $F_1 = 1$ and $F_{n+1} = F_n + F_{n-1}$. Let $M = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$.

(a) Show by induction that $M^n = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix}$ for $n \geq 1$, and compute M^{10}.

(b) Use the product rule for determinants to prove Cassini's identity $F_{n+1}F_{n-1} - F_n^2 = (-1)^n$. Check it for $n = 10$.

Problem 6.29. (a) Find 2×2 matrices A and B with $AB - BA = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. Why is there no such pair with $AB - BA = I_2$?

(b) Show that if $\operatorname{tr}(A^\top A) = 0$, then $A = 0$.

(c) Let A be an $n \times n$ matrix with $\operatorname{tr}(AX) = 0$ for every $n \times n$ matrix X. Show that $A = 0$. (Hint: try $X = e_{ji}$, the matrix with a single 1 in position (j, i).)

Problem 6.30 ($\star$). (a) How many of the 16 matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with all entries in $\{0, 1\}$ are invertible?

(b) Show that a square matrix A with integer entries has an inverse with integer entries if and only if $\det(A) = \pm 1$.

6.6 Block Matrices, the LU Decomposition and the Woodbury Identity

Problem 6.31 (Mandatory). Let $A = \begin{pmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{pmatrix}$.

(a) Find the LU decomposition $A = LU$ by Gaussian elimination, recording the multipliers in L.

(b) Use it to compute $\det(A)$, and to solve $A\mathbf{x} = (2, 10, 3)^\top$ by forward and back substitution.

Problem 6.32. Show that $A = \begin{pmatrix} 0 & 2 \\ 3 & 1 \end{pmatrix}$ has no factorization $A = LU$ with L unit lower triangular and U upper triangular. Find a permutation matrix P such that $PA = LU$, and give L and U.

Problem 6.33 (Mandatory). Let $M = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$, split into the blocks $A = (2)$, $B = (1, 1)$, $C = B^\top$, and the lower right 2×2 block D.

(a) Compute the Schur complement S of A and $\det(M) = \det(A)\det(S)$.

(b) Compute S^{-1}, which is the lower right 2×2 block of M^{-1}. Then find all of M^{-1} with the formula of the lecture, and check that $MM^{-1} = I_3$.

Problem 6.34. Let $A = \operatorname{diag}(1, 2, 4)$ and $\mathbf{u} = \mathbf{v} = (1, 1, 1)^\top$. Use the Sherman–Morrison formula to compute $(A + \mathbf{u}\mathbf{v}^\top)^{-1}$, and the determinant lemma to compute $\det(A + \mathbf{u}\mathbf{v}^\top)$. Check your answer for the determinant by row reduction.

Problem 6.35 ($\star$)**.** Let J_n be the $n \times n$ matrix whose entries are all 1, so that $J_n = \mathbf{1}\mathbf{1}^\top$ for the vector $\mathbf{1}$ of ones.

(a) Show that $\det(aI_n + bJ_n) = a^{n-1}(a + nb)$ for all numbers $a \neq 0$ and b.

(b) For which numbers x is the $n \times n$ matrix with all diagonal entries equal to x and all other entries equal to 1 singular?

(c) For $a \neq 0$ and $a + nb \neq 0$, find $(aI_n + bJ_n)^{-1}$. (It is again of the form $\alpha I_n + \beta J_n$.)

Problem 6.36. Let X be an $N \times d$ matrix and $\lambda > 0$.

(a) Show that $(X^\top X + \lambda I_d)^{-1} X^\top = X^\top (XX^\top + \lambda I_N)^{-1}$.

(b) For a single data point, $X = \mathbf{x}^\top$ with $\mathbf{x} \in \mathbb{R}^d$, use (a) to show that $(\mathbf{x}\mathbf{x}^\top + \lambda I_d)^{-1}\mathbf{x} = \frac{1}{\mathbf{x}^\top \mathbf{x} + \lambda}\,\mathbf{x}$. Check this directly by multiplying both sides by $\mathbf{x}\mathbf{x}^\top + \lambda I_d$.

7 Null Space, Column Space and Rank

Problem 7.1 (**Mandatory**)**.** Let $B = \begin{pmatrix} 1 & 2 & 0 & 3 \\ 2 & 4 & 1 & 8 \\ 1 & 2 & 1 & 5 \end{pmatrix}$. Find $\mathrm{RREF}(B)$, the rank of B, and bases of $\mathrm{null}(B)$, $\mathrm{col}(B)$ and $\mathrm{row}(B)$. Check that your answer agrees with the rank–nullity theorem.

Problem 7.2 (**Mandatory**, $\star$)**.** (a) Let $T : \mathbb{R}^5 \to \mathbb{R}^3$ be a surjective linear transformation. What is the dimension of $\ker(T)$?

(b) Can a linear transformation $\mathbb{R}^3 \to \mathbb{R}^5$ be surjective? Can it be injective?

(c) Can a 4×6 matrix A have $\mathrm{null}(A) = \{\mathbf{0}\}$?

(d) Let A be a 5×3 matrix such that $A\mathbf{x} = \mathbf{0}$ only for $\mathbf{x} = \mathbf{0}$. What is the rank of A?

Problem 7.3 ($\star$)**.** (a) Let V and W be 4-dimensional subspaces of $\mathbb{R}^7$. Show that $V \cap W \neq \{0\}$. (Hint: put a basis of V and a basis of W as the columns of a 7×8 matrix, and apply the rank–nullity theorem.)

(b) More generally, show that if V and W are subspaces of $\mathbb{R}^n$ with $\dim V + \dim W > n$, then $V \cap W \neq \{0\}$.

(c) Find two 3-dimensional subspaces of $\mathbb{R}^6$ whose intersection is $\{0\}$.

Problem 7.4. (a) Show that $\mathrm{rank}(AB) \leq \mathrm{rank}(A)$ and $\mathrm{rank}(AB) \leq \mathrm{rank}(B)$ whenever AB is defined. (Hint: $\mathrm{col}(AB) \subseteq \mathrm{col}(A)$ and $\mathrm{null}(B) \subseteq \mathrm{null}(AB)$.)

(b) Conclude that if A is 3×2 and B is 2×3, then the 3×3 matrix AB is never invertible.

Problem 7.5. Let A be an $m \times n$ matrix.

(a) Show that $\mathrm{null}(A^\top A) = \mathrm{null}(A)$. (Hint: if $A^\top A\mathbf{x} = \mathbf{0}$, then $\mathbf{x}^\top A^\top A\mathbf{x} = 0$; what is $\mathbf{x}^\top A^\top A\mathbf{x}$ in terms of the entries of $A\mathbf{x}$?)

(b) Conclude that $\mathrm{rank}(A^\top A) = \mathrm{rank}(A)$.

(c) Show that if the columns of A are linearly independent, then $A^\top A$ is invertible. (This is why the normal equations $(A^\top A)\mathbf{w} = A^\top \mathbf{y}$ of linear regression have a unique solution.)

8 Dual Vector Spaces

Problem 8.1 (Mandatory)**.** Which of the following maps are linear functionals? For those on $\mathbb{R}^n$, write them as $\mathbf{x} \mapsto \mathbf{a}\mathbf{x}$ with a row vector $\mathbf{a}$.

(a) $\mathbb{R}^3 \to \mathbb{R}$, $\mathbf{x} \mapsto 2x_1 - x_3$;

(b) $\mathbb{R}^3 \to \mathbb{R}$, $\mathbf{x} \mapsto x_1 x_2$;

(c) $\mathbb{R}^{2\times 2} \to \mathbb{R}$, $A \mapsto \operatorname{tr}(A)$;

(d) $\mathbb{R}^{2\times 2} \to \mathbb{R}$, $A \mapsto \det(A)$;

(e) $\mathbb{R}[x]_{\le 3} \to \mathbb{R}$, $p \mapsto p(2) - p'(0)$;

(f) $\mathbb{R}[x]_{\le 3} \to \mathbb{R}$, $p \mapsto p(0)^2$.

Problem 8.2 (Mandatory)**.** (a) Find the dual basis b^1, b^2 of the basis $\mathbf{b}_1 = (2,1)^\top$, $\mathbf{b}_2 = (1,1)^\top$ of $\mathbb{R}^2$, and check that $b^i(\mathbf{b}_j) = \delta_{ij}$.

(b) Write the functional $\varphi(\mathbf{x}) = 3x_1 + x_2$ as a linear combination of b^1 and b^2, without solving a system of equations.

Problem 8.3 ($\star$)**.** Let $V = \mathbb{R}[x]_{\le 2}$ and consider the nodes 0, 1, 2.

(a) Find the Lagrange polynomials $L_0, L_1, L_2 \in V$ with $L_j(k) = \delta_{jk}$ for $j, k \in \{0,1,2\}$.

(b) Write $p(x) = x^2 + 1$ as a linear combination of L_0, L_1, L_2.

(c) The functional $I(p) = \int_0^2 p(x)\,dx$ on V is a linear combination of the evaluation functionals ev_0, ev_1, ev_2. Find the coefficients by computing $I(L_0)$, $I(L_1)$ and $I(L_2)$. (You have discovered Simpson's rule for numerical integration, which is exact for polynomials of degree at most 2.)

Problem 8.4 (Mandatory)**.** Let $T : \mathbb{R}^3 \to \mathbb{R}^2$, $T(x,y,z) = (x + y,\ y - z)$, with matrix A.

(a) For the functional ψ on $\mathbb{R}^2$ given by the row $(2, -1)$, compute $T^*(\psi)$ as a row vector.

(b) Check that $T^*(\psi)(\mathbf{v}) = \psi(T(\mathbf{v}))$ for $\mathbf{v} = (1,2,3)^\top$.

(c) Find the matrix of T^* with respect to the duals of the standard bases, and compare it with A.

Problem 8.5. (a) Find a basis of the annihilator of $U = \operatorname{span}\{(1,2,0)^\top, (0,1,1)^\top\}$ in $(\mathbb{R}^3)^*$, and check that $\dim U + \dim U^\circ = 3$.

(b) Find a basis of the annihilator of $W = \operatorname{span}\{(1,0,1,0)^\top, (0,1,0,1)^\top\}$ in $(\mathbb{R}^4)^*$.

Problem 8.6 ($\star$)**.** (a) Show that the system $x_1 + 2x_2 = 1$, $2x_1 + 4x_2 = 3$ has no solution by finding a row $\mathbf{y}$ with $\mathbf{y}A = \mathbf{0}$ and $\mathbf{y}\mathbf{b} \neq 0$.

(b) For which $\mathbf{b} = (b_1, b_2, b_3)^\top$ does the system with the matrix $A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{pmatrix}$ have a solution? Answer by finding all rows $\mathbf{y}$ with $\mathbf{y}A = \mathbf{0}$ and using the Fredholm alternative.

Problem 8.7. Let $V = \mathbb{R}^2$, and identify V^* with row vectors. Describe the element $\iota(\mathbf{e}_1) \in V^{**}$, i.e., compute $\iota(\mathbf{e}_1)(\varphi)$ for the functional φ given by a row (a_1, a_2). Then show directly that $\iota(v) = 0$ implies $v = 0$.

9 Quotient Spaces and the Isomorphism Theorems

Problem 9.1 (**Mandatory**)**.** Let $U = \operatorname{span}\{(2,1)^\top\} \subseteq \mathbb{R}^2$.

(a) Decide whether $(3,1)^\top + U = (1,0)^\top + U$, whether $(5,3)^\top + U = (1,1)^\top + U$, and whether $(1,2)^\top + U = U$.

(b) Show that the cosets of U are the lines $x_1 - 2x_2 = c$, and find the unique representative of each coset on the x_1-axis.

(c) Find an explicit isomorphism $\mathbb{R}^2/U \to \mathbb{R}$.

Problem 9.2. Find the dimensions of the following quotient spaces, and describe each of them as a familiar vector space:

(a) $\mathbb{R}^5/U$, where $\dim U = 2$;

(b) $\mathbb{R}^{3\times 3}/\mathrm{Sym}_3$;

(c) $\mathbb{R}[x]_{\le 4}/U$, where $U = \{p : p(1) = p(2) = 0\}$. (Hint: consider $p \mapsto (p(1), p(2))$.)

Problem 9.3 (**Mandatory**)**.** Use the first isomorphism theorem to identify the following quotient spaces.

(a) $\mathbb{R}^3/P$, where P is the plane $x + y + z = 0$. (Use $T(x,y,z) = x + y + z$.)

(b) $\mathbb{R}[x]_{\le 3}/U$, where $U = \{p : p(0) = p(1) = 0\}$. What is $\dim U$?

(c) $\mathbb{R}^{n\times n}/\mathrm{Sym}_n$. (Use $T(A) = A - A^\top$, and find its kernel and its image.)

Problem 9.4 (**Mandatory**, $\star$)**.** (a) Let V and W be 4-dimensional subspaces of a 7-dimensional vector space. Which values can $\dim(V \cap W)$ take? Give an example for the smallest possible value.

(b) Let $U = \{\mathbf{x} \in \mathbb{R}^4 : x_1 + x_2 + x_3 + x_4 = 0\}$ and $W = \{\mathbf{x} \in \mathbb{R}^4 : x_1 - x_2 + x_3 - x_4 = 0\}$. Use the dimension formula to find $\dim(U \cap W)$, and then find a basis of $U \cap W$.

Problem 9.5. Let $V = \mathbb{R}^4$, $U = \operatorname{span}\{\mathbf{e}_4\}$ and $W = \operatorname{span}\{\mathbf{e}_3, \mathbf{e}_4\}$. Describe V/U, W/U and V/W, and check the third isomorphism theorem by comparing dimensions.

Problem 9.6 (**Mandatory**, $\star$)**.** The softmax function is $\operatorname{softmax}(\mathbf{z})_i = \frac{e^{z_i}}{\sum_j e^{z_j}}$ for $\mathbf{z} \in \mathbb{R}^n$.

(a) Show that $\operatorname{softmax}(\mathbf{z} + c\mathbf{1}) = \operatorname{softmax}(\mathbf{z})$ for every $c \in \mathbb{R}$, so that softmax is a function on $\mathbb{R}^n/\operatorname{span}\{\mathbf{1}\}$.

(b) Compute $\operatorname{softmax}(1,2,3)$ and $\operatorname{softmax}(1001, 1002, 1003)$. Why can a computer evaluate the second one only after subtracting a constant from all entries?

(c) Is softmax injective on $\mathbb{R}^n/\operatorname{span}\{\mathbf{1}\}$? (Hint: if $\operatorname{softmax}(\mathbf{z}) = \operatorname{softmax}(\mathbf{w})$, compare $z_i - z_j$ and $w_i - w_j$.)

Problem 9.7. Let $U = \operatorname{span}\{(1,1,1)^\top\} \subseteq \mathbb{R}^3$. Describe all linear functionals on $\mathbb{R}^3/U$ as rows $\mathbf{a} = (a_1, a_2, a_3)$, and check that they form a 2-dimensional space, in agreement with $(V/U)^* \cong U^\circ$.

10 Eigenvalues and Eigenvectors

Problem 10.1 (**Mandatory**)**.** Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}$.

(a) Compute the characteristic polynomial of A, its eigenvalues, and its eigenspaces.

(b) Check that the eigenvalues add up to $\operatorname{tr}(A)$ and multiply to $\det(A)$.

(c) Diagonalize A, i.e., find P and D with $A = PDP^{-1}$.

(d) Find a formula for A^k, and check it for $k = 1$.

Problem 10.2. Find the eigenvalues and eigenspaces of $L = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 3 & 0 \\ 4 & -1 & 1 \end{pmatrix}$. Is L diagonalizable?

Problem 10.3 (**Mandatory**)**.** Let $N = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$. Find the algebraic and geometric multiplicity of each eigenvalue of N, and decide whether N is diagonalizable.

Problem 10.4. Without computing any characteristic polynomial, use geometric intuition to find the eigenvalues and eigenspaces of the following linear transformations of the plane. Then check your answers with a computation.

(a) the reflection in the x-axis;

(b) the orthogonal projection onto the line $y = x$, with matrix $\frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$;

(c) the shear $T(x, y) = (x + 2y, y)$;

(d) the rotation by 90°;

(e) $T(x, y) = (2x, 3y)$;

(f) $T(x, y) = (-x, -y)$.

Which of these transformations are diagonalizable?

Problem 10.5 ($\star$)**.** A real 3×3 matrix A has the eigenvalues -1, 2 and 4.

(a) Is A diagonalizable? Is A invertible?

(b) Compute $\operatorname{tr}(A)$, $\det(A)$, $\operatorname{tr}(A^2)$ and $\det(A^3)$.

(c) Find the eigenvalues of A^{-1}, A^2 and $A + 2I_3$.

Problem 10.6 ($\star$)**.** Let P be an $n \times n$ matrix with $P^2 = P$ (compare Problem 6.13).

(a) Show that every eigenvalue of P is 0 or 1.

(b) Show that every nonzero vector of $\operatorname{im}(T_P)$ is an eigenvector for the eigenvalue 1, and that every nonzero vector of $\ker(T_P)$ is an eigenvector for the eigenvalue 0. Use Problem 6.13(d) to conclude that P is diagonalizable.

(c) Which of the following statements must be true: P is invertible; P is the identity or the zero matrix; P is diagonalizable?

Problem 10.7. Let A and B be $n \times n$ matrices.

(a) Show that A and $A^\top$ have the same eigenvalues, and give an example where they have different eigenvectors.

(b) Show that if A is invertible, then AB and BA are similar, and hence have the same eigenvalues. (Hint: $BA = A^{-1}(AB)A$.)

(c) Show that every real eigenvalue of an orthogonal matrix Q is 1 or -1. Hint: if $Q\mathbf{v} = \lambda\mathbf{v}$, compute $(Q\mathbf{v})^\top(Q\mathbf{v})$ in two ways.

(d) Show that if $A^k = 0$ for some $k \geq 1$, then 0 is the only eigenvalue of A.

(e) Show that $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and I_2 have the same characteristic polynomial, but are not similar.

Problem 10.8 ($\star$)**.** The sequence $a_0 = 0$, $a_1 = 1$, $a_{n+1} = a_n + 2a_{n-1}$ starts with $0, 1, 1, 3, 5, 11, \ldots$.

(a) Find a 2×2 matrix M with $(a_{n+1}, a_n)^\top = M(a_n, a_{n-1})^\top$.

(b) Find the eigenvalues and eigenvectors of M, and derive a formula for a_n, as for Binet's formula in the lecture.

(c) Compute a_{10} with your formula.

Problem 10.9 (**Mandatory**)**.** Every day, a visitor of a website either only browses or makes a purchase. A browsing visitor browses again the next day with probability 0.7, and a buying visitor buys again with probability 0.6. With $x_k = (b_k, p_k)^\top$ the probabilities of browsing and buying on day k, we have $x_{k+1} = Ax_k$ with

$$A = \begin{pmatrix} 0.7 & 0.4 \\ 0.3 & 0.6 \end{pmatrix}.$$

(a) Find the eigenvalues and eigenvectors of A, and the stationary distribution.

(b) For $x_0 = (1, 0)^\top$, find a formula for x_k, and check it for $k = 2$. What happens as $k \to \infty$?

Problem 10.10. Let $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ and $x_0 = (1, 0)^\top$.

(a) Compute Ax_0, A^2x_0, A^3x_0 and A^4x_0. What happens to the ratio of the two entries?

(b) Explain your observation by writing x_0 as a linear combination of eigenvectors of A, and find a formula for A^kx_0.

Problem 10.11 ($\star$)**.** (a) Show that the matrix $M(a + bi) = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$ of Problem 6.5 has the complex eigenvalues $a \pm bi$.

(b) Find the complex eigenvalues of $\begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix}$, and write this matrix as rR_θ with $r > 0$ and a rotation R_θ. How are r and θ related to the eigenvalues?

Problem 10.12 ($\star$)**.** Let $D : C^\infty(\mathbb{R}) \to C^\infty(\mathbb{R})$, $Df = f'$. As for matrices, we call $f \neq 0$ an eigenvector of D with the eigenvalue λ if $Df = \lambda f$.

(a) Show that every real number λ is an eigenvalue of D, with the eigenvector $e^{\lambda x}$, and that every eigenvector for λ is a multiple of $e^{\lambda x}$. (Hint: differentiate $f(x)e^{-\lambda x}$.)

(b) Show that e^x, e^{2x} and e^{3x} are linearly independent, without any computation. (Hint: the theorem that eigenvectors for different eigenvalues are linearly independent holds for every linear transformation $V \to V$, with the same proof.)

(c) Conclude in a second way that $C^\infty(\mathbb{R})$ is infinite-dimensional.

Problem 10.13 (**Mandatory**)**.** Let $B = \begin{pmatrix} 1 & 1 \\ -2 & 4 \end{pmatrix}$.

(a) Compute p_B, and verify the Cayley–Hamilton theorem by computing $p_B(B)$.

(b) Write B^{-1} and B^3 in the form $aB + bI_2$.

(c) Show that $B^k = a_k B + b_k I_2$ for every $k \geq 1$, with $a_{k+1} = 5a_k + b_k$ and $b_{k+1} = -6a_k$.

Problem 10.14. Find the minimal polynomials of the following matrices, and decide which of them are diagonalizable:

$$N = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad D = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{pmatrix}, \qquad R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Without computing any eigenvectors, show that a matrix A with $A^2 = 3A - 2I_n$ is diagonalizable, and find its possible eigenvalues.

Problem 10.15 (**Mandatory**)**.** Let $n = 4$, $\mathbf{c} = (1, 1, 0, 0)^\top$ and $\mathbf{x} = (1, 0, 2, 0)^\top$, with entries numbered from 0, so that $(\mathbf{c} * \mathbf{x})_j = x_j + x_{j-1}$.

(a) Write down $\mathrm{circ}(\mathbf{c})$, and compute $\mathbf{c} * \mathbf{x}$.

(b) Compute the DFTs $\hat{\mathbf{c}}$, $\hat{\mathbf{x}}$ and $\widehat{\mathbf{c} * \mathbf{x}}$ (here $\omega = i$), and check the convolution theorem.

(c) Which eigenvalue of $\mathrm{circ}(\mathbf{c})$ is 0, and for which eigenvector? Explain the answer without computation.

Problem 10.16 ($\star$)**.** Let S be the 3×3 cyclic shift, $S(x_0, x_1, x_2)^\top = (x_2, x_0, x_1)^\top$. Find all 3×3 matrices A with $AS = SA$ by solving the linear equations for the entries of A, and check that they are exactly the circulant matrices.

11 Tensors

Problem 11.1 (**Mandatory**)**.** (a) Use the bilinearity of $(v, w) \mapsto v \otimes w$ to show that $(2v) \otimes w = v \otimes (2w)$ and $v \otimes w + v \otimes (-w) = 0$ in $V \otimes W$.

(b) Expand $(\mathbf{e}_1 + 2\mathbf{e}_2) \otimes (3\mathbf{e}_1 - \mathbf{e}_2) \in \mathbb{R}^2 \otimes \mathbb{R}^2$ in the basis $\mathbf{e}_i \otimes \mathbf{e}_j$, and write it as a 2×2 matrix.

(c) Which of the tensors $\mathbf{e}_1 \otimes \mathbf{e}_1 + \mathbf{e}_1 \otimes \mathbf{e}_2 + \mathbf{e}_2 \otimes \mathbf{e}_1 + \mathbf{e}_2 \otimes \mathbf{e}_2$ and $\mathbf{e}_1 \otimes \mathbf{e}_2 - \mathbf{e}_2 \otimes \mathbf{e}_1$ in $\mathbb{R}^2 \otimes \mathbb{R}^2$ are pure tensors?

(d) What is the dimension of $\mathbb{R}^2 \otimes \mathbb{R}^3 \otimes \mathbb{R}^4$, and of $(\mathbb{R}^2 \otimes \mathbb{R}^3)^*$?

Problem 11.2 (**Mandatory**). (a) A video has 30 color frames per second, each of 64×64 pixels with 3 color channels. What are the order and the shape of the array that stores a 10-second video? Write the space of all such arrays as a tensor product $\mathbb{R}^{n_1} \otimes \cdots \otimes \mathbb{R}^{n_k}$, and find its dimension.

(b) A tensor $T \in \mathbb{R}^{2\times 3\times 4}$ is flattened row by row (with the last index changing fastest) into a vector $\mathbf{v} \in \mathbb{R}^{24}$ with the entries $v_1, \ldots, v_{24}$. Which entry of $\mathbf{v}$ is T_{213}? Which entry of T is v_{17}?

Problem 11.3 (**Mandatory**). Which of the following maps $\mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$ are bilinear? For the bilinear ones, find the matrix A with $\beta(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top A \mathbf{y}$.

(a) $\beta(\mathbf{x}, \mathbf{y}) = 2x_1y_1 - x_1y_2 - x_2y_1 + 4x_2y_2$;

(b) $\beta(\mathbf{x}, \mathbf{y}) = x_1y_1 + 1$;

(c) $\beta(\mathbf{x}, \mathbf{y}) = (x_1 - x_2)(y_1 + 2y_2)$;

(d) $\beta(\mathbf{x}, \mathbf{y}) = x_1^2 y_2$.

Which of the bilinear ones satisfy $\beta(\mathbf{x}, \mathbf{y}) = \beta(\mathbf{y}, \mathbf{x})$ for all $\mathbf{x}$ and $\mathbf{y}$?

Problem 11.4 (**Mandatory**). Let $\mathbf{u} = (1, -1)^\top$, $\mathbf{v} = (2, 0, 1)^\top$ and $\mathbf{w} = (1, 3)^\top$.

(a) Write the pure tensor $\mathbf{u} \otimes \mathbf{v} \in \mathbb{R}^2 \otimes \mathbb{R}^3$ as a matrix, and find its rank.

(b) Compute the coordinates of $T = \mathbf{u} \otimes \mathbf{v} \otimes \mathbf{w} \in \mathbb{R}^2 \otimes \mathbb{R}^3 \otimes \mathbb{R}^2$, and write down its two slices $(T_{ij1})_{i,j}$ and $(T_{ij2})_{i,j}$.

(c) Is the matrix $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ an outer product $\mathbf{a} \otimes \mathbf{b}$? Write it as a sum of as few outer products as possible.

Problem 11.5 (**Mandatory**). For each of the following expressions, give the Einstein notation and the corresponding call of the NumPy function `np.einsum`, and state the shape of the result.

(a) $A^\top \mathbf{x}$ for $A \in \mathbb{R}^{m\times n}$ and $\mathbf{x} \in \mathbb{R}^m$;

(b) the sum of all entries of the entrywise product $A \odot B$, where $(A \odot B)_{ij} = A_{ij}B_{ij}$;

(c) the diagonal of AB for $n \times n$ matrices A and B, without computing all of AB; how many multiplications does it need?

(d) the outputs $XW^\top$ of a linear layer, for a batch X of shape $N \times d$ and a weight matrix $W \in \mathbb{R}^{m\times d}$.

Problem 11.6. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$, $T(x, y) = (x + 2y, 3y)$.

(a) Write the tensor in $\mathbb{R}^2 \otimes (\mathbb{R}^2)^*$ that corresponds to T in the basis $\mathbf{e}_i \otimes e^j$, and as a sum of two pure tensors.

(b) Compute its contraction, and compare it with the trace of the matrix of T.

(c) Compute the matrix of T in the basis $b_1 = (1, 1)$, $b_2 = (1, -1)$, and check that it has the same trace.

Problem 11.7. Let $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$, and let $P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, whose columns form a basis of $\mathbb{R}^2$.

(a) Compute the matrix $P^{-1}AP$ of the linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ in this basis.

(b) Compute the matrix $P^{\top}AP$ of the bilinear form $(\mathbf{x}, \mathbf{y}) \mapsto \mathbf{x}^{\top}A\mathbf{y}$ in this basis.

(c) Compare the traces and the determinants of A, $P^{-1}AP$ and $P^{\top}AP$. Which of them depend on whether A describes a linear transformation or a bilinear form?

Problem 11.8 ($\star$)**.** (a) Let $T \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ be a sum of r rank-one tensors. Show that its unfolding, the $n_1 \times (n_2 n_3)$ matrix whose i-th row lists the entries T_{ijk}, has rank at most r.

(b) Use (a) to show again that the tensor $\mathbf{e}_1 \otimes \mathbf{e}_1 \otimes \mathbf{e}_1 + \mathbf{e}_2 \otimes \mathbf{e}_2 \otimes \mathbf{e}_2$ of the lecture has rank 2.

(c) Show that the tensor $T \in \mathbb{R}^{2 \times 2 \times 2}$ with $T_{ijk} = 1$ for all i, j, k has rank one.